

Osmania University

REGULAR EXAMINATION ? 2025

Electromagnetic Theory and Transmission Lines ? Regular Examination, 2025

Subject Code: EC404 Semester: 4 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Electrostatics)

1. Describe the working principle of Electrostatics with necessary equations. [12 Marks]
2. Write short notes on Electrostatics. [8 Marks]
3. Solve a problem related to Electrostatics. Show all the steps clearly. [12 Marks]

SECTION B (Unit 2: Magnetostatics)

4. Discuss the engineering applications of Magnetostatics. [12 Marks]
5. Solve a problem related to Magnetostatics. Show all the steps clearly. [12 Marks]
6. Solve a problem related to Magnetostatics. Show all the steps clearly. [12 Marks]

SECTION C (Unit 3: Maxwell's Equations)

7. Explain Maxwell's Equations with a neat diagram/sketch and a suitable example. [12 Marks]
8. Solve a problem related to Maxwell's Equations. Show all the steps clearly. [12 Marks]
9. Write short notes on Maxwell's Equations. [12 Marks]

SECTION D (Unit 4: Electromagnetic Wave Propagation)

10. Discuss the engineering applications of Electromagnetic Wave Propagation. [8 Marks]
11. Describe the working principle of Electromagnetic Wave Propagation with necessary equations. [12 Marks]
12. Compare and contrast the key concepts involved in Electromagnetic Wave Propagation. [12 Marks]

SECTION E (Unit 5: Transmission Lines)

13. Describe the working principle of Transmission Lines with necessary equations. [8 Marks]
14. Explain Transmission Lines with a neat diagram/sketch and a suitable example. [12 Marks]
15. Solve a problem related to Transmission Lines. Show all the steps clearly. [12 Marks]

*** End of Question Paper ***